Nathaniel C. Moll

Lynchburg, VA | 610-463-4974 | ncmoll45@gmail.com | Citizenship: United States

Summary: Biochemist and multidisciplinary researcher with hands-on experience in molecular biology, protein purification, quantitative biochemical assays, and DNA biophysics. Experienced in experimental design, protocol optimization, and computational analysis of biological data. Interested in leveraging AI-assisted scientific workflows to accelerate biological discovery and advance human health.

Core Competencies: Molecular Biology & Biochemistry (PCR, gel electrophoresis, SDS-PAGE, protein purification, bacterial cell culture, CRISPR-Cas9) • Biophysical & Organic Chemistry (DLS, SEM, NMR/IR, polymer synthesis) • Bioinformatics & Machine Learning (Python/TensorFlow, BLAST, YOLO, Spiking Neural Networks) • Multi-Sensor & Data Fusion (LiDAR, Stereo-Depth, Neuromorphic) • Engineering Design (SolidWorks CAD, ASME Y14.5)

RESEARCH & WORK EXPERIENCE

Molecularly Imprinted Polymer Synthesis for PFAS Detection **08/2025 – Present**

Liberty University | Lynchburg, VA | Supervisor: Dr. Stephen Hobson, sthobson@liberty.edu, May Be Contacted

Relevant Experience: Designed fluorinated molecularly imprinted polymers (fMIPs) to selectively bind PFAS "forever chemicals" and organophosphate pesticides, synthesizing and characterizing novel compounds for high-sensitivity environmental sensors.

Polymer Synthesis Team Lead **08/2026 – Present**

15 hrs/wk, Volunteer

- Directed research tasks among other students, trained younger students on reaction methods, and presented weekly progress updates to the faculty mentor and team.
- Utilized AI tools for research optimization such as Claude for developing formal reaction protocols, compiling information from the literature to inform operating procedures, and troubleshooting chemical extraction and reaction yield issues.
- Poor reaction yields warranted investigation into a new procedure. Researched alternative reaction methods and performed reaction in new chemical conditions, bringing product yield up by 25%.
- Directed setup and execution of new reactions, achieving high product purity and setting precedents for future students.

Research Team Member **08/2025 – 05/2026**

15 hrs/wk, Volunteer

- Many organic reactions were time-consuming and product-poor. Optimized extraction and wash-cycle protocols for imprinted polymer synthesis procedures, raising reaction yield from baseline to as high as 75%.
- Reaction products contained impurities and needed to be separated. Designed a flash-chromatography setup to remove residual water and contaminants, improving product purity for high-sensitivity PFAS sensor testing.
- Verified novel molecular structures via IR and NMR spectroscopy; presented findings at the Virginia Academy of Science Annual Meeting and secured a $1,000 grant to expand the work.

Science Undergraduate Laboratory Internship (SULI), DOE Research Intern **05/2026 – 07/2026**

Oak Ridge National Laboratory | Oak Ridge, TN | 40 hrs/wk | Supervisor: Dr. Daniel Archer, archerde@ornl.gov, May Be Contacted

Relevant Experience: Advanced Detection and Applied Data Science group (Physics Division): applied data systems, sensor workflows, and cyber-physical systems to roadside capture nodes that detect nuclear source material in moving vehicles.

- Designed and built an end-to-end LiDAR data systems pipeline (Python) that isolated moving vehicles from a static background and generated annotated datasets for training a LiDAR object detection model.
- Deployed and trained a YOLO computer vision model to benchmark LiDAR, stereo, and neuromorphic sensor modalities, reaching 87% detection performance on the first training pass across 17,000+ 2D-projected LiDAR frames.
- Performed cyber-physical operations (networking, power, data acquisition, sensor timestamp synchronization) to troubleshoot and maintain each capture node, ensuring readiness for scene-accurate field data collection.

Physical Chemistry Research Experience for Undergraduates (REU) **05/2025 – 07/2025**

University of Mississippi | Oxford, MS | 40 hrs/wk | Supervisor: Dr. Randy Wadkins, rwadkins@olemiss.edu, May Be Contacted

Relevant Experience: Investigated mercury-ion-induced nanoparticle formation with short DNA fragments, designing and executing experiments to characterize binding dynamics and nanoparticle morphology for biosensor applications.

- Independently designed spectrofluorometric and DLS assays, requiring benchtop formulation and incubation of mercury–DNA samples.
- Size distribution data of T-rich DNA nanoparticles were noisy and complex. Performed Dynamic Light Scattering (DLS) assays across 500+ mercury-DNA samples and engineered a custom unsupervised machine-learning pipeline (Python/TensorFlow) to resolve signals.
- Biophysical dynamics of mercury-induced nanoparticles were understudied. Determined a novel 3:1 mercury:DNA binding ratio via spectrofluorometric assay (K_d = 14.4 µM; cooperativity = 2.81), resolving a gap in the existing literature.

- Designed and executed an SEM imaging protocol to directly visualize DNA nanoparticle morphology, confirming ML-derived size-distribution conclusions.
- Implemented statistical algorithms in Python such as PCA and data clustering algorithms to visualize DNA sample data.
- Delivered a comprehensive 15-minute oral capstone presentation to 30+ faculty and students.

NASA Human Exploration Rover Challenge (HERC), Suspension Sub-team Lead **05/2024 – 05/2025**

Liberty University School of Engineering | Lynchburg, VA | 15 hrs/wk, Volunteer | Supervisor: Zach Huff, 434-851-1475, May Be Contacted

- Led a rover suspension from design through fabrication for a NASA competition team: developed a new geometry delivering 7" of total wheel travel, modeled it in SolidWorks with ASME Y14.5-compliant drawings, and coordinated systems integration across sub-teams.
- Presented Operational Readiness Reviews to NASA engineers to validate design progress.

Engineering Research & Development Intern **05/2024 – 08/2024**

United Safety and Survivability Corporation | Exton, PA | 25 hrs/wk | Supervisor: Matthew Riley, mriley@usscgroup.com, May Be Contacted

- Ran high-frequency shake-table tests to verify oscillation-damping specs on industrial seat suspensions; designed a new air-canister mounting bracket that cut a fabrication bottleneck and reduced seat weight.
- Redesigned a portable fire-suppression demo unit in SolidWorks under tight dimensional constraints, producing ASME Y14.5-compliant drawings for every part to guide prototype fabrication.

EDUCATION

Liberty University | Lynchburg, VA **Expected 05/2027**

B.S., Biochemistry & Molecular Biology; Minor in Physics | GPA: 3.96/4.0 | Honors College, Dean's List

Relevant Coursework: Calculus I–III, Physics I & II, Mechanics, Electricity & Magnetism, Organic Chemistry I & II, Analytical Chemistry, Physical Chemistry, Biochemistry I & II, Genetics, Cell Biology, Microbiology, Bioinformatics, Medicinal Chemistry

Wet-Laboratory Projects (hands-on techniques):

- **Genetics:** DNA extraction and purification, PCR, gel electrophoresis, plasmid design, restriction endonuclease assays.
- **Microbiology:** aseptic technique, bacterial cell culture and stable line maintenance; identified unknown bacteria via PCR of 16S rRNA gene and gel electrophoresis.
- **Cell Biology:** lab research project on the mitochondrial release mechanism of cytochrome c, using differential centrifugation, protein purification, SDS-PAGE, and enzyme activity assays to investigate mode of release.
- **Biochemistry:** lab research project isolating peroxidases from novel sources such as pumpkin and zucchini. Protein purification (via SEC) and enzyme activity assays used to assess concentration of peroxidases in the various plant tissue locations. Tissue staining and histological analysis also performed.
- **Bioinformatics:** sequence alignment (BLAST), sequence homology analysis, primer design, and wet-lab procedure involving bacterial transformation with CRISPR-Cas9 for gene disruption and knock-out. Cas9 crRNA analyzed for off-target effects.
- **Medicinal Chemistry:** molecular docking in-silico experiment to inspect active-site amino acid configuration with a small molecule bound.

Honors Senior Research Thesis: Designing a neuromorphic spiking neural network to classify cell-free DNA nanopore signals and detect tumor fractions non-invasively from blood samples. Progress to date: dataset construction and model architecture configuration. Collaborating with computer scientists from Oak Ridge National Laboratory.

PUBLICATIONS AND PRESENTATIONS

- Moll, N. "Multipartite Synthesis of Molecularly Imprinted Polymers for PFAS Detection." Poster presentation, Virginia Academy of Science Annual Meeting, Virginia State University, 2025.
- Moll, N. "The Effects of Mercury Ions on Secondary Structures Formed by T-Rich DNA." Oral presentation, NSF Research Experience for Undergraduates, University of Mississippi, 2025. Manuscript in preparation for journal submission in collaboration with Dr. Randy Wadkins.

ADDITIONAL INFORMATION

Leadership, Memberships and Volunteering: Founder & President, Liberty University Chemistry Club (ACS Student Chapter), 03/2024–present; Student Member, American Chemical Society; Volunteer Operator, Liberty University Astronomical Observatory, 03/2024–present.

Awards & Honors: Virginia Academy of Science Research Grant, $1,000 (2025); Full-Tuition Merit Scholarship, Liberty University (2023–2027); Honors College Member; Dean's List (multiple semesters); Reading Musical Foundation Scholarship, classical piano, 4× awardee (2019–2023).

Technical Skills: PCR, gel electrophoresis, SDS-PAGE, protein purification, SEC, enzyme kinetic assays, cell culture, CRISPR-Cas9, BLAST, primer design, DNA/RNA extraction, molecular docking, NMR/IR, DLS, SEM, spectrofluorometry, column chromatography, organic synthesis • Python (TensorFlow, scikit-learn, pandas, NumPy, SciPy, Open3D), HPC, SolidWorks CAD